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Simulation of magnetic field line stochasticity at the magnetopause

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Abstract. We have conducted a three-dimensional particle simulation to study the magnetic field line stochasticity at the magnetopause current layer. Our results show that the magnetic field lines become stochastic due to the overlap of the destabilized multiple tearing mode islands, which agrees with the percolation model suggested by Galeev et al. (1986). After the field lines become stochastic, these tearing modes grow even 2-3 times faster than in the linear stage and saturate at an amplitude level 3-4 times bigger than the single tearing mode without mode-mode coupling. The field line stochasticity also causes a strong particle diffusion across the current layer. The diffusion coefficient reaches to 10^9 m^2/s for typical magnetopause parameters. Associated with the particle diffusion, the current layer becomes broader in width. As a result, the magnetic energy is dissipated into particle energy by heating parallel to the local magnetic field. The particle energy increases by 60%, while the magnetic helicity, which has always been regarded as a good invariant, changes by 20%.

1. Introduction

The transport of solar wind energy, momentum, and particles through the magnetopause boundary is crucial to determine the magnetospheric dynamics. There are two major processes to account for the transport: the viscous (or diffusive) process [*Axford and Hines*, 1961] and the magnetic reconnection process [*Dungey*, 1961]. In the viscous interaction process, the solar wind energy and momentum are transported into the magnetosphere by a viscous stress at the magnetopause boundary. The viscous and diffusive effects are also important to the formation of the low-latitude boundary layer at the magnetopause [*Sonnerup*, 1980]. It was suggested that a viscosity or diffusion coefficient of 10^9 m^2/s is needed to maintain the boundary layer. Obviously, for a collisionless plasma, such a diffusion coefficient can only be achieved through anomalous processes due to wave-particle and wave-wave interactions [*LaBelle and Treumann*, 1988]. In the magnetic reconnection process, the solar wind energy and momentum are transported directly into the open magnetosphere through the magnetic reconnection. More importantly, this latter process causes the solar wind convection electric field to map down to the Earth's ionosphere and therefore create the circulation of the magnetosphere. The magnetic reconnection process at the magnetopause boundary has been studied extensively [*Levy et al.*, 1964; *Russell and Elphic*, 1978; *Lee and Fu*, 1985; *Galeev et al.*, 1986].

One of the major models for the reconnection is the percolation model proposed by *Galeev et al.* [1986]. In this model, it was suggested that the reconnection might be induced by multiple tearing modes growing on different magnetic surfaces, and the overlap of these tearing mode islands could cause field lines inside the magnetopause current layer to be stochastic. Therefore a field line starting from the magnetosheath side could connect to the magnetospheric side via an irregular path. Because of its similarity to water percolating through a random medium, it was named the percolation model. In fact, the transport process in the percolation model is more or less in between the classical reconnection type and the diffusive type. In this model, the magnetic field lines do connect to the magnetospheric side from the solar wind. However, the solar wind convection electric field cannot be mapped down to the ionosphere due to the randomness of the magnetic field lines. Thus some kind of coherent process is needed to drive the circulation of the magnetosphere. On the other hand, since the field lines connecting the solar wind and the magnetosphere are stochastic, we expect that the diffusion of particles is very strong. In fact, the diffusion coefficient does reach to 10^9 m^2/s when calculated from quasi-linear theory. Galeev et al. proposed their model based on two major considerations, i.e., the linear theory of tearing mode instability and the kinematic study of field line stochasticity. In the first part, they performed a linear theory study on the tearing mode. In particular, they studied the growth rate dependence on the magnetic field configuration and concluded that an angle of 60° shear was the best for the tearing mode onset on different magnetic surfaces. In the second part, they used quasilinear theory to calculate the field line diffusion coefficient, which has then been used to estimate particle diffusion coefficients.

In addition to Galeev et al.'s work, there are many other papers on these two subjects. The collisionless tearing mode instability has been studied extensively in many situations [*Coppi et al.*, 1966; *Schindler and Soop*, 1968; *Drake and Lee*, 1977; *Coroniti*, 1980; *Quest and Coroniti*, 1981; *Gladd*,

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1990]. The study of field line stochasticity has also been studied extensively in other plasmas, such as controlled fusion and astrophysics [*Rosenbluth et al.*, 1966; *Jokipii and Parker*, 1969; *White*, 1983; *Tetreault*, 1988]. In most of the studies of field line stochasticity, the problems are approached by the kinematics of static magnetic field lines. In this approach, the topological properties of the field lines are studied under a certain specified magnetic field, typically with broadband spatial fluctuations. The field line diffusion coefficient, which represents the randomness of the magnetic field lines, is then calculated from quasi-linear theory [*Rosenbluth et al.*, 1966; *Galeev et al.*, 1986]. The onset of stochasticity as the result of spatial perturbations on the magnetic field has also been studied [*Rechester and White*, 1980; *White*, 1983]. However, the self-consistent aspect of the field line stochasticity that describes the plasma response to the stochastic field line and the evolution of the field line stochasticity in a plasma is mostly unaddressed. Certain questions, such as the relaxation of the magnetic field, the heating of particles as a result of dissipation of magnetic energy, and the influence on the plasma instabilities (e.g., tearing mode) after the onset of field line stochasticity still remain unanswered or only partially answered. *Tetreault* [1988, 1989, 1992a, b] has tried to answer the relaxation problem in the framework of magnetohydrodynamics (MHD). He suggests that Taylor state [*Taylor*, 1974], which is a minimum magnetic energy state under the constraint of invariant magnetic helicity, can be approached by the MHD clump instability. Such an instability is generated under the condition of stochastic magnetic field lines. In a recent work, *Galeev* [1991] suggests that the stochastic field line condition will enhance the growth rate of the tearing mode.

In this work, we carry out a numerical particle simulation to study the stochastic field line problem in a self-consistent manner. We solve the first principle equations (equation of motion for particles and Maxwell equations for the electromagnetic fields) numerically in a stochastic magnetic field situation. The system we simulate is a Harris equilibrium with a guide field. After initializing such a system, we let the system evolve by itself by solving the first principle equations. The tearing modes which are unstable in this system will then grow and cause the field line to be stochastic. The evolution of the system will then be determined only by the initial conditions.

Since the field line stochasticity only happens in a system with three-dimensional spatial dependence (a two-dimensional system corresponds to a one-dimensional Hamiltonian system, which is always integrable), we have to perform the simulation in a three-dimensional system. Due to the limitation on the computer capability, we perform this three-dimensional electromagnetic particle simulation using an eigenmode expansion method [*Cheng and Okuda*, 1977] instead of the widely used particle-in-cell method. Like the spectral method in some fluid codes, the eigenmode expansion method uses Fourier representation on field variables. By choosing a limited number of modes, we can perform the simulation under present computer capability and thus simulate the stochastic field line problem with the coupling of a limited number of modes. From our simulation, we find that the tearing modes grow from an initial thermal fluctuation level in a modified Harris equilibrium and then cause the field lines to be stochastic as the tearing mode islands grow to sufficient amplitude. As the field lines become stochastic, particles diffuse across the current layer quickly and are heated significantly along the local magnetic field. The tearing modes which are responsible for the field line stochasticity grow 2-3 times faster and saturate at an amplitude 3-4 times greater than that in the linear stage. The magnetic helicity is an absolute invariant for the ideal MHD and is also regarded as a good invariant for the relaxation of a magnetic system even after the field line stochasticity [*Taylor*, 1974; *Tetreault*, 1988]. However, we have calculated the magnetic helicity numerically and find that the magnetic helicity changes by 20% as the particle energy increases by 60%.

This paper is organized as following. In section 2, we describe the numerical method for the three-dimensional particle simulation. Section 3 presents the numerical results. Section 4 includes the discussion and the conclusions.

2. Numerical Method

To attack this problem, we apply a three-dimensional electromagnetic particle simulation. Instead of using the widely used particle-in-cell method, we use the eigenmode expansion method [*Cheng and Okuda*, 1977], which deals with field quantities in Fourier space. We first generalize the method from its electrostatic version [*Cheng and Okuda*, 1977] to an electromagnetic version. The detailed description on the eigenmode expansion method is given by *Cheng and Okuda* [1977]. The basics of this method is to push the particles in real configuration space and then obtain the charge density and current density directly in Fourier space instead of collecting it on the grid first. After obtaining the electromagnetic fields in Fourier space, the fields are imposed on the particles by mode summation instead of doing a fast Fourier transform back to the grid.

The geometry we use is shown in Figure 1. The magnetopause is chosen to be in a slab geometry because the scale length we are interested in ($O(R_E)$ is much smaller than the radius of curvature of the magnetopause ($O(10\ R_E)$). The magnetopause normal is in the x direction toward the Sun. The initial magnetic field is in the yz plane, in which we choose a constant B_y across the magnetopause and a B_z with a hyperbolic tangent dependence on x. This magnetic field configuration, together with the particle distribution,

$$f_s = n_o\left(\frac{m_s}{2\pi T_s}\right)^{\frac{3}{2}}\exp\left(\frac{-m(v_x^2 + (v_y - v_{ds})^2 + v_z^2)}{2T_s}\right)\mathrm{sech}^2\left(\frac{x}{L}\right)$$

constitutes the so-called modified Harris equilibrium [*Harris*, 1962; *Galeev et al.*, 1986].

The fundamental equations for a collisionless plasma are the Vlasov equation and Maxwell equations. In particle simulations, the Vlasov equation is solved by integrating its characteristic equations, which are just the equations of motion for the particles. The charge density and the current density are then obtained by accumulating particles either on a grid or in Fourier modes. The Maxwell equations are then solved with proper boundary conditions. The equations we solve are

$$f(\mathbf{x}, \mathbf{v}, t) = \sum_j^N \delta(\mathbf{x} - \mathbf{x}_j(t))\delta(\mathbf{v} - \mathbf{v}_j(t)) \tag{1}$$

$$\frac{d\mathbf{x}_j(t)}{dt} = \mathbf{v}_j(t) \tag{2}$$

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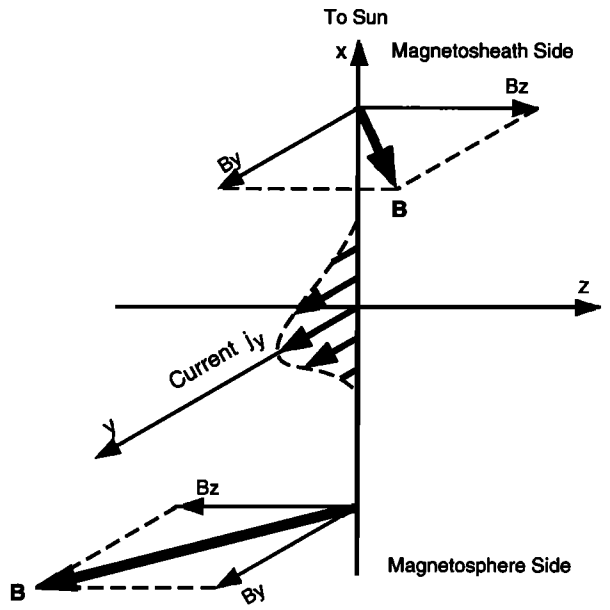


Fig. 1. Schematic of the magnetopause current layer. The magnetopause normal is along the x direction (toward the Sun). The initial magnetic field is in the yz plane with constant B_y and a B_z with a hyperbolic tangent dependence on x across the current layer. The magnetopause current is in the y direction. In our simulation, the parameters are chosen such that the field has a 90° shear and the shear scale length is 4 ion Larmor radii.

$$\frac{d\mathbf{v}_j(t)}{dt} = \frac{q_s}{m_s}(\mathbf{E} + \frac{\mathbf{v}_j}{c}\times\mathbf{B}) \tag{3}$$

$$\rho_s(\mathbf{x},t) = \sum_j^N \delta(\mathbf{x} - \mathbf{x}_j(t)) \tag{4}$$

$$\mathbf{j}_s(\mathbf{x},t) = \sum_j^N \mathbf{v}_j(t)\delta(\mathbf{x} - \mathbf{x}_j(t)) \tag{5}$$

$$\nabla\cdot\mathbf{E} = 4\pi\sum_s q_s\rho_s(\mathbf{x},t) \tag{6}$$

$$\nabla\times\mathbf{B} = \frac{1}{c}\frac{\partial\mathbf{E}}{\partial t} + \frac{4\pi}{c}\sum_s \mathbf{j}_s \tag{7}$$

$$\nabla\times\mathbf{E} = -\frac{1}{c}\frac{\partial\mathbf{B}}{\partial t} \tag{8}$$

$$\nabla\cdot\mathbf{B} = 0$$

We assume that the boundary conditions in y and z directions are periodic. The wavenumbers of the Fourier modes are $k_y = 2\pi m/L_y$ and $k_z = 2\pi n/L_z$, where m and n are limited in a small range $|m| < M$ and $|n| < N$. After performing a Fourier transform on Maxwell equations, we get

$$\frac{\partial^2}{\partial x^2}\Phi(x,k_y,k_z) - (k_y^2 + k_z^2)\Phi(x,k_y,k_z) = 4\pi\sum_s q_s\rho_s(x,k_y,k_z) \tag{9}$$

$$\mathbf{E}_l(x,k_y,k_z) = -\nabla_{k_y,k_z}\Phi(x,k_y,k_z) \tag{10}$$

for the electrostatic (longitudinal) part. The electromagnetic (transverse) part becomes

$$\frac{\partial}{\partial t}\mathbf{B}(x,k_y,k_z) = -\nabla_{k_y,k_z}\times\mathbf{E}_t \tag{11}$$

$$\frac{\partial}{\partial t}\mathbf{E}_t(x,k_y,k_z) = \nabla_{k_y,k_z}\times\mathbf{B} - 4\pi\mathbf{j}_t(x,k_y,k_z) \tag{12}$$

where

$$\nabla_{k_y,k_z} \equiv (\frac{\partial}{\partial x}, ik_y, ik_z)$$

Our next step is to obtain the Fourier-transformed current density and charge density $\rho(x,k_y,k_z)$ and $j_t(x,k_y,k_z)$. In the particle-in-cell method, this is carried out by loading particles on a three-dimensional grid, first using equations (4) and (5) and then performing a numerical fast Fourier transform (FFT). Since we do not use a grid in the y and z directions in the eigenmode expansion method, we perform an analytic Fourier transform on equations (4) and (5) first, and then obtain the Fourier-transformed charge density and current density directly from

$$\rho_s(x,k_y,k_z) = \sum_j^N \delta(x - x_j)e^{\imath k_y y_j + \imath k_z z_j} \tag{13}$$

$$\vec{j}_s(x,k_y,k_z) = \sum_j^N \vec{v}_j\delta(x - x_j)e^{\imath k_y y_j + \imath k_z z_j} \tag{14}$$

After we obtain the electric and magnetic fields from the Fourier mode expression $\mathbf{E}(x,k_y,k_z)$ in (9)-(12), we apply them directly to the particle locations by summing up the modes, i.e.,

$$\mathbf{E}(x_j,y_j,z_j) = \sum_{m=-M,n=-N}^{M,N} \mathbf{E}(x_j,k_y,k_z)e^{-\imath k_y y_j - \imath k_z z_j} \tag{15}$$

$$\mathbf{B}(x_j,y_j,z_j) = \sum_{m=-M,n=-N}^{M,N} \mathbf{B}(x_j,k_y,k_z)e^{-\imath k_y y_j - \imath k_z z_j} \tag{16}$$

This avoids the use of three-dimensional grids in the particle-in-cell method. In the x direction we still get the field by linearly interpolating from the values on the grid, which means we replace the δ function in (13) and (14) with a triangle-shaped function.

The implementation on time integral in equations (1)-(7) is chosen to be exactly the same as in the conventional electromagnetic particle-in-cell code [e.g., *Pritchett*, 1985], i.e., a leapfrog scheme for the particle pusher and a leapfrog on Ampere's law and Faraday's law.

To further simplify the numerical calculation on Maxwell equations (9) – (12), we double the system in the x direction and put a mirror image of the system with reversed current direction in the extra box. This gives a periodic boundary in the x direction in the doubled system. This operation will, of course, change the simulation object slightly from a single current layer, shown in Figure 2*a*, into a series of current layers with opposite sign which are periodically spaced in the x direction, shown in Figure 2*b*. However, as long as the simulation box is much bigger than the width of the current layer, it will not alter the results significantly, because the

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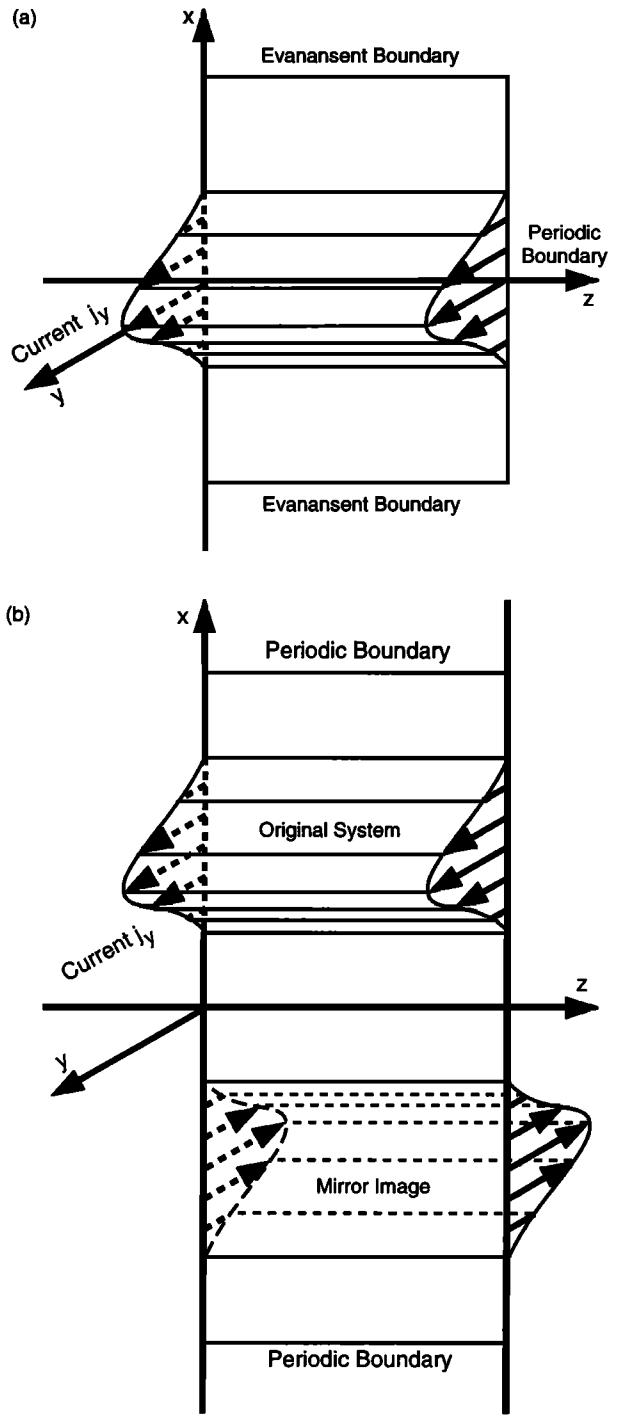


Fig. 2. Schematic of our simulation system. (*a*) The system that we intend to simulate. The current sheet is centered at $x = 0$. The boundary conditions are periodic in the y and z directions and evanescent in the x direction. (*b*) The system we use in the simulation. For the sake of checking the conservation laws, the evanescent boundary condition in the x direction is avoided in this system, in which our original system is doubled by putting a mirror image in the system. The doubled system has a periodic boundary condition in the x direction.

tearing mode is evanescently decaying away from the current layer. The advantage of using this modified system instead of the original one is that the total energy is conserved and therefore can be used as a diagnostic.

3. Numerical Results

A modified Harris equilibrium,

$$f_s = n_o\left(\frac{m_s}{2\pi T_s}\right)^{\frac{3}{2}}\exp\left(\frac{-m(v_x^2+(v_y-v_{ds})^2+v_z^2)}{2T_s}\right)\mathrm{sech}^2\left(\frac{x}{L}\right)$$

is initialized with 20,000 electrons and 20,000 ions, where $v_{ds} = 2cT_s/q_sB_{zo}L$. The magnetic field configuration corresponding to this distribution is $B_y = B_{yo}$ and $B_z = B_{zo}\tanh(x/L)$. Figure 3 shows spatial variations of a few physical variables (B_z, B_y, j_y, and n) corresponding to this equilibrium. For a more realistic magnetopause equilibrium, particles should also be loaded to both the magnetosheath side and the magnetospheric side. The particles we have here are the trapped particle component used in the magnetopause equilibrium modeling [*Lee and Kan*, 1979]. The numerical simulation is then performed under the parametric conditions in Table 1. From Table 1 we can see that certain parameters are close to realistic magnetopause conditions such as β, L/R_i, and $L/c/\omega_{pi}$, where β is the ratio between the thermal pressure and the magnetic pressure, L is the magnetopause scale length, R_i is the ion Larmor radius, and c/ω_{pi} is the ion inertia length. But some are 1 order of magnitude different from the realistic magnetopause value, such as L/λ_e, where λ_e is the Debye length. Such a difference is inevitable in most particle simulations. We argue that this difference would not significantly change our explanation of the simulation results. For example, for the problem that the Debye length is bigger in the simulation than in the real magnetopause, we argue that the Debye length reflects the scale length within which the charge neutrality holds. In a tearing mode, charge neutrality should hold within a region with a scale length similar to a wavelength. Since the wavelength is still much bigger than the Debye length ($O(10)$), we believe the relatively big Debye length will not significantly alter the results. Another

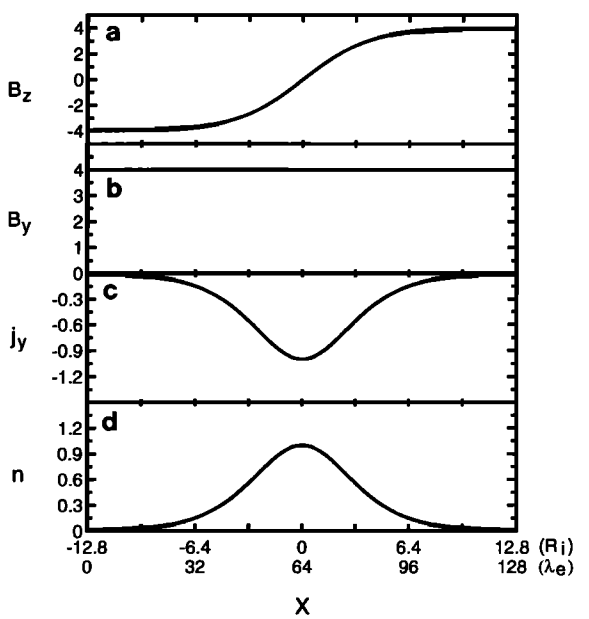


Fig. 3. Initial conditions of the simulation (a modified Harris equilibrium). (*a*) The magnetic field B_z component. It has a hyperbolic tangent dependence on x. (*b*) The magnetic field B_y component. It is a constant across the current layer. (*c*) The current j_y. Initially j_z is zero. (*d*) Density profile.

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TABLE 1. Simulation Parameters

Names	Symbols	Values
Mass ratio	m_e/m_i	0.25
Debye length	λ_{pe} (x=x_{center})	1
Current scale length	L	$20\lambda_{pe}$ =4R_i
Ion thermal speed	V_{Ti}	1
Electron thermal speed	V_{Te}	2
Alfven speed	V_A (x=x_{center})	2V_{Ti}
Speed of light	c	5
Magnetic field y component	B_y	4
Magnetic field z component	B_{zo} (x=x_{center})	4

issue concerns the mass ratio. As is well known, electrons should dominate the dissipation current in the linear stage of the tearing mode because of their small inertia [e.g., *Quest and Coroniti*, 1981]. If the electron mass becomes as big as that of ions, the dissipation will include both electrons and ions. Since we have chosen a mass ratio of 1/4, the electron dissipation current will still be 4 times bigger than that of ions. Therefore the relatively small mass ratio will not cause confusion between the ion tearing and electron tearing modes in the linear stage. However, for the single tearing mode, the saturation level of the tearing mode strongly depends on the mass of the resonant particles. This is because the tearing mode grows until the resonant particles are trapped in the islands. Since the electrons travel much faster than the ions, the electron tearing mode reaches saturation much more quickly than ion tearing. The electron tearing mode therefore has a much lower saturation level than the ion tearing mode. Since we increase electron mass to one fourth that of ions in this simulation, the saturation level of the electron tearing mode will be increased to a level comparable to the ion tearing mode, which causes the separation of electron tearing mode from ion tearing mode to be difficult at the nonlinear stage. Certain nonlinear aspects, such as that the ion tearing mode will take over after the electron tearing mode saturates [*Kuznetsova and Zelenyi*, 1990], will be difficult to resolve in the current parameter set. For the case when there are multiple tearing modes, particles will not be trapped if the islands are destroyed as a result of their overlap with one another. The saturation mechanism in this case is different from the single tearing mode case. The role that mass ratio plays in the destroyed magnetic surfaces situation is not clear.

We perform our simulation in two steps. First, we study cases in which an individual tearing mode is isolated from others. We do so by allowing only one specified mode to exist in the simulation system. We then study the case with mode-mode interaction in the system by allowing several modes to coexist. This two-step study enables us to understand the mode-mode coupling effect in great detail, as shown in the field line stochasticity.

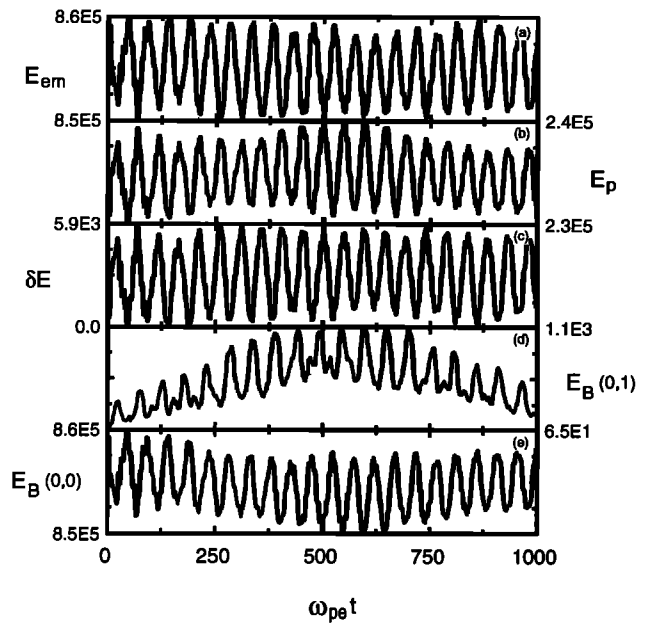
3.1 Isolated Mode With $k_y = 0$ and $k_z = 2\pi/L_z$

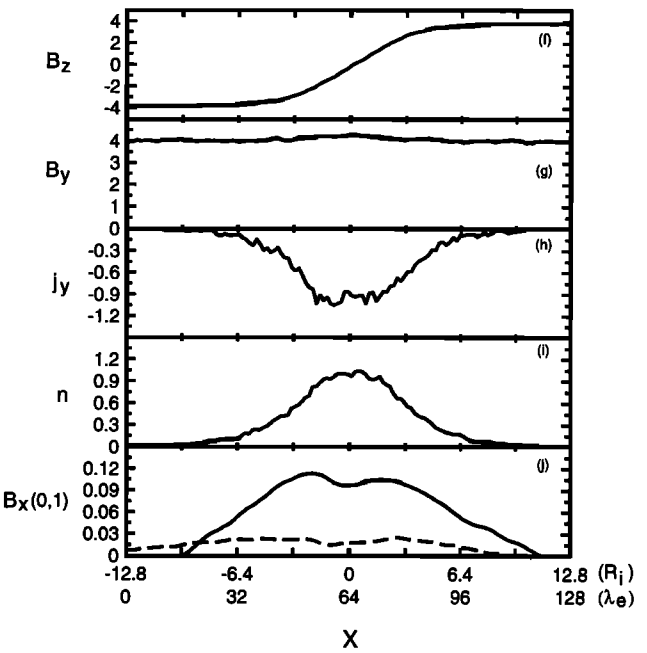
Figure 4 illustrates the result from our first step, the case of an isolated tearing mode with mode number $m = 0$ and $n = 0, \pm 1$. Note that mode (m =0, n =-1) is the complex conjugate of the mode (m =0, n =+1) which guarantees a real value for field quantities. This choice of mode number allows a single tearing mode ($m = 0, n = 1$) with wavenumber $k_y = 0$ and $k_z = 2\pi/L_z$ to grow. Figure 4*a* shows the time variation of the electromagnetic energy of the whole system. Overall, the electromagnetic energy does not change significantly (of the order of 1%). Looking at the overall trend, there is a decrease until $T = 500$, ($T \equiv \omega_{pe}t$) and then a slight increase. The most obvious signature in this panel is a fast oscillation with a period of 50. This oscillation is not related to the tearing mode phenomena that we are interested in. It is a MHD magnetosonic mode oscillation resulting from the slightly unbalanced initial loading. Figure 4*b* shows the time variation of the total particle energy. Again there is no significant particle energy gain or loss. Since the total energy should be conserved in this system, the loss of the electromagnetic energy should be equal to the gain of the particle energy. This is illustrated in Figure 4*c*, which shows the electromagnetic energy loss $-(E_{em}(t) - E_{em}(0))$ and the particle energy gain $E_p(t) - E_p(0)$. The two curves nearly exactly coincide with each other. Actually, the final energy deviates less than 0.01% from the initial energy (i.e., $[E_{tot}(t) - E_{tot}(0)]/E_{tot}(0) < 10^{-4}$). The following two panels show the magnetic energy, i.e., $E_B(m,n) = (1/2)\int \mathbf{B}(x,m,n)^2 dx$ for each mode. Figure 4*d* shows the magnetic energy for the tearing mode ($m = 0, n = 1$). The magnetic energy of the tearing mode increases initially and saturates at T=500, and then decreases with time. In Figure 4*e*, the magnetic energy of mode (m =0, n =0), or the background magnetic energy, decreases with time.

In the next five panels of Figure 4, we show snapshots of the magnetic field, density, and current at T =500, when the tearing mode saturates. Figure 4*f* shows the magnetic field B_z as a function of x. We can see that the hyperbolic tangent magnetic field profile has not been changed significantly. Figure 4*g* shows that the B_y component increases only slightly in the center of the current layer. The current distribution, shown in Figure 4*h*, is slightly modified in the center of the current layer, where a slight flattening on the current appears. In fact, this is expected from the linear tearing mode theory. The free energy term for the tearing mode is associated with a second-order derivative on the background magnetic field [*Schindler*, 1966; *Wang and Ashour-Abdalla*, 1992], i.e., first-order derivative on the current. If the current flattens, there is no more free energy available to the tearing mode. Thus it will saturate. The flattening area corresponds to the area of the magnetic island, which will be described in Figure 4*k*. Figure 4*i* shows that the density is also slightly flattened at the center of the current layer, though it is not as obvious as the current. The profile of the normal component B_x of the magnetic field for the tearing mode is displayed in Figure 4*j*, with the solid line for the imaginary part and the dashed line for the real part. The structure of the mode is very much the same as that obtained analytically from linear tearing mode theory [e.g., *Kuznetsova and Zeleny*, 1985], where $B_x = B_{xo}P_1^{k_z L}(x)$ is a Legendre polynomial. We have calculated the growth rate for the mode ($m = 0, n = 1$) in the interval from T=0 to T=500, and we find that the linear growth rate is $\gamma = 0.002\omega_{pe}$. This result is in good agreement with the linear theory estimation $\gamma/kV_{Te} = \Delta'/\sqrt{\pi}l_s k_{oe}^2$ (see *Quest and Coroniti* [1981] for details), which gives $\gamma = 0.003\omega_{pe}$ when using the parameters given in Table 1.

In Figure 4*k*, magnetic field lines for the tearing mode at $T = 500$ are projected on the xz plane. The plot has been generated as a surface of section plot (see Appendix B for

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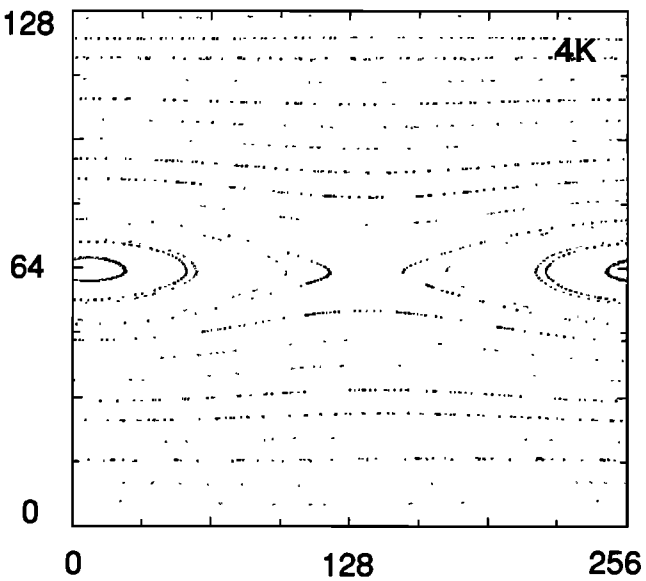


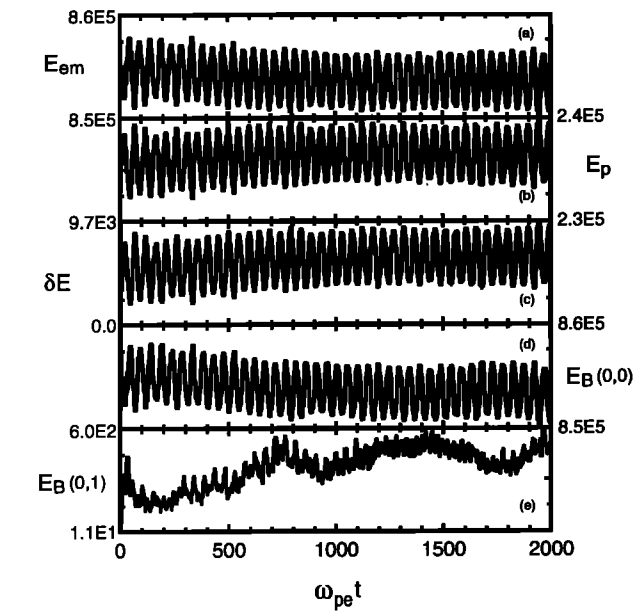
Fig. 4. Simulation results for the case with an isolated tearing mode ($k_y = 0$ and $k_z = 2\pi/L_z$) showing the time variations of energies and snapshots at $T = 500$. (*a*) The time variation of total electromagnetic energy of the system. (*b*) The time variation of total particle energy. (*c*) The time variations of the particle energy gain and the electromagnetic energy loss. The two curves coincide, indicating the total energy is conserved in the simulation. (*d*) The time variation of the magnetic field energy for the tearing mode ($m = 0, n = 1$). (*e*) The time variation of the background magnetic energy, or mode ($m = 0, n = 0$). (*f*) The profile of the magnetic field B_z component at $T = 500$. (*g*) The profile of the magnetic field B_y component at $T = 500$. (*h*) The profile of the current j_y at $T = 500$. (*i*) The profile of density at $T = 500$. (*j*) The profile of the magnetic normal component B_x for mode ($m = 0, n = 1$) at $T = 500$. The solid and dashed traces correspond to the imaginary and real parts, respectively. (*k*) The surface of section plot in the xy plane at $T = 500$. A magnetic island is formed in the center of the current layer.

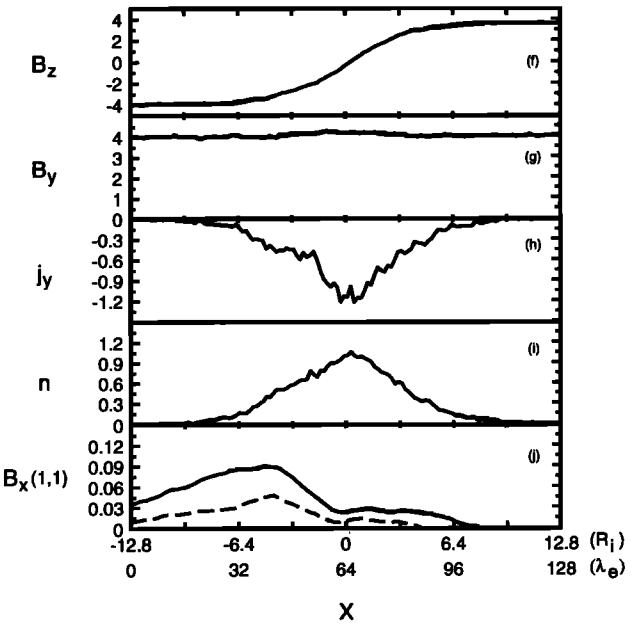
details). It shows clearly that a magnetic island has developed at the center of the current layer. The width of the current layer is about $20\lambda_e$, which is the same as the scale length of the current layer.

3.2. *Isolated Mode With $k_y = 2\pi/L_y$ and $k_z = 2\pi/L_z$*

Figure 5 shows a series of plots for the mode number ($m = 1$, $n = 1$; $m = -1$, $n = -1$; $m = 0$, $n = 0$), where the mode ($m = 0, n = 0$) is for the background field and the mode ($m = -1, n = -1$) is the complex conjugate of the mode ($m = 1, n = 1$) to guarantee real values on field quantities. This choice of mode number allows only a single tearing mode with wavenumber $k_y = 2\pi/L_y, k_z = 2\pi/L_z$ to grow. From the linear tearing mode theory [*Galeev et al.*, 1986], we know that the tearing mode islands grow on the resonance surface where $\mathbf{k} \bullet \mathbf{B} = 0$. Calculating from this equation, we get the location of the resonant surface $x = x_c = 54\lambda_e$ for the simulation. Figure 5*a* shows the time variation of the electromagnetic energy for the whole system. Again there is no significant overall energy change (about 1%). There is a trend, although not obvious, that the average energy decreases with time until $T = 700$, and then saturates. Similar to the previous case, a fast magnetosonic oscillation also occurs due to the slightly unbalanced initial loading. The particle energy does not increase significantly either, as shown in Figure 5*b*. The loss of electromagnetic energy and the increase of the particle energy are equal to each other in Figure 5*c*, where two curves coincide. The total energy conservation in our simulation is again better than 0.01%. Figure 5*d* shows the time variation of the magnetic energy for the background field $E_B(m = 0, n = 0)$, which decreases with time until $T = 700$ and then stays at the same level. Figure 5*e* shows the magnetic energy of tearing mode $E_B(m = 1, n = 1)$ increases with time until $T = 700$ and then becomes saturated.

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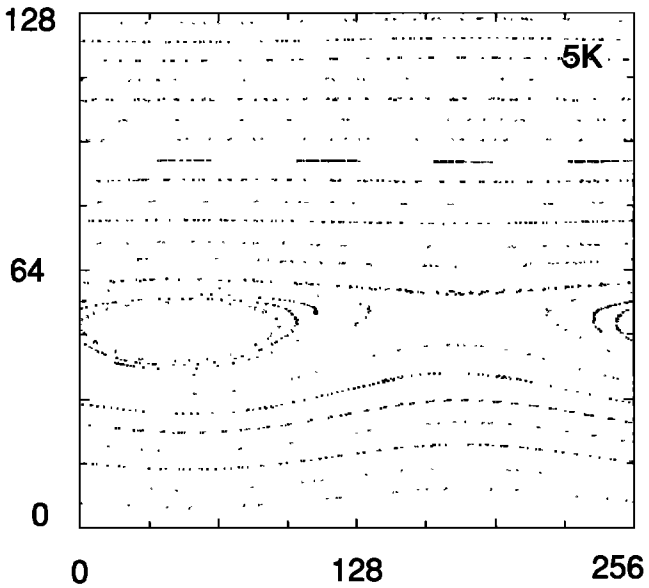


Fig. 5. Simulation results for the case with an isolated tearing mode ($k_y = 2\pi/L_y$ and $k_z = 2\pi/L_z$) showing the time variations of energies and snapshots at $T = 1000$. (*a*) The time variation of total electromagnetic energy of the system. (*b*) The time variation of total particle energy. (*c*) The time variations of the particle energy gain and the electromagnetic energy loss. The two curves coincide, indicating the total energy is conserved in the simulation. (*d*) The time variation of the background magnetic energy, or mode ($m = 0, n = 0$). (*e*) The time variation of the magnetic field energy for the tearing mode ($m = 1, n = 1$). (*f*) The profile of the magnetic field B_z component at $T = 1000$. (*g*) The profile of the magnetic field B_y component at $T = 1000$. (*h*) The profile of the current j_y at $T = 1000$. (*i*) The profile of density at $T = 1000$. (*j*) The profile of the magnetic normal component B_x for mode ($m = 1, n = 1$) at $T = 1000$. The solid and dashed traces correspond to the imaginary and real parts, respectively. (*k*) The surface of section plot in the xy plane at $T = 1000$. A magnetic island is formed at the side of the current layer.

We now show snapshots of the density, current, and magnetic field at $T = 1000$, when the tearing mode saturates. Figures 5*f* and 5*g* show the magnetic field components B_z and B_y as a function of x, respectively. Again there is little change in the magnetic field. However, we can see obvious changes in the current profile (Figure 5*h*) and density profile (Figure 5*i*). The current is flattened around $x \approx 48\lambda_e$. The flattening area corresponds to the area of the magnetic island, which will be shown in Figure 5*k*. As we mentioned above, the free energy term in the linear tearing mode theory requires a first-order derivative on the current. A flattening on the current means an exhaustion of the free energy. Therefore the tearing mode will saturate. Figure 5*j* shows the profile of magnetic normal component $B_x(x, m = 1, n = 1)$ for mode ($m = 1, n = 1$). The mode is peaked at $x \approx 40$ which is on the left side of the resonant surface $x = 54\lambda_e$. There is also a dip in the center of the current layer $x = 64\lambda_e$. In Figure 5*k*, we present a plot showing the magnetic islands for the tearing mode. The plot again is generated as a surface of section plot (see detailed description in Appendix B). It can be understood as the intersection between flux tubes and the xz plane. It shows a magnetic island formed at the side of the current layer. In fact, the reconnection X point is located at the resonance surface $x = 54\lambda_e$. The width of the island in the x direction is about $20\lambda_e$ which is the scale length of the current.

3.3. *Multiple Tearing Modes*

We now examine the case in which multiple modes are allowed to grow together. The parameters are still the same as shown in Table 1. The modes are $m = -3, -2, -1, 0, 1, 2, 3$ and $n = -1, 0, 1$.

Figure 6*a* shows the time variation of the total electromagnetic energy. In contrast to the cases with single tearing mode, it shows significant decrease of the electromagnetic energy with time, especially after $T = 750$.

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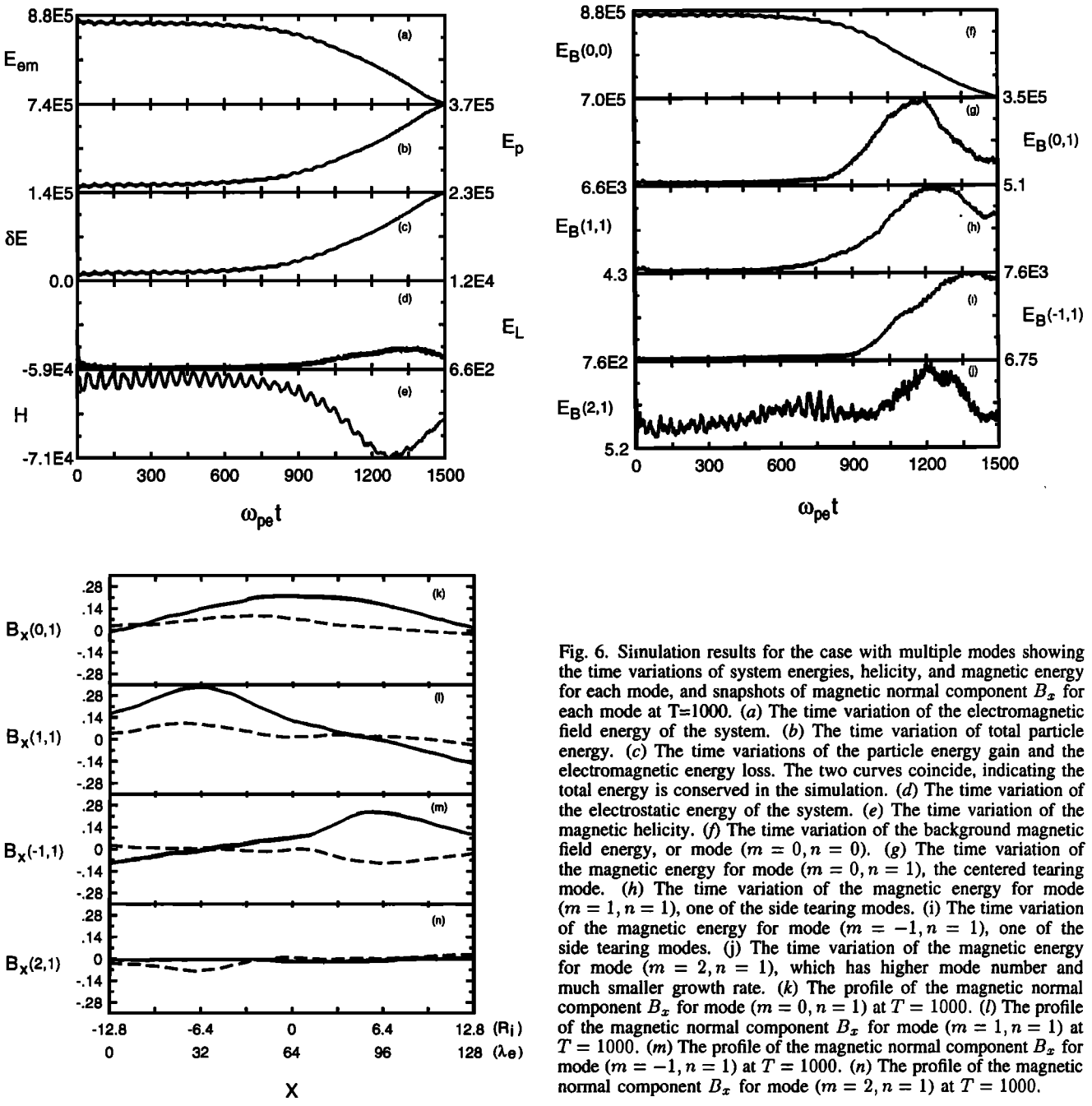


Fig. 6. Simulation results for the case with multiple modes showing the time variations of system energies, helicity, and magnetic energy for each mode, and snapshots of magnetic normal component B_x for each mode at T=1000. (*a*) The time variation of the electromagnetic field energy of the system. (*b*) The time variation of total particle energy. (*c*) The time variations of the particle energy gain and the electromagnetic energy loss. The two curves coincide, indicating the total energy is conserved in the simulation. (*d*) The time variation of the electrostatic energy of the system. (*e*) The time variation of the magnetic helicity. (*f*) The time variation of the background magnetic field energy, or mode ($m = 0, n = 0$). (*g*) The time variation of the magnetic energy for mode ($m = 0, n = 1$), the centered tearing mode. (*h*) The time variation of the magnetic energy for mode ($m = 1, n = 1$), one of the side tearing modes. (i) The time variation of the magnetic energy for mode ($m = -1, n = 1$), one of the side tearing modes. (j) The time variation of the magnetic energy for mode ($m = 2, n = 1$), which has higher mode number and much smaller growth rate. (*k*) The profile of the magnetic normal component B_x for mode ($m = 0, n = 1$) at $T = 1000$. (*l*) The profile of the magnetic normal component B_x for mode ($m = 1, n = 1$) at $T = 1000$. (*m*) The profile of the magnetic normal component B_x for mode ($m = -1, n = 1$) at $T = 1000$. (*n*) The profile of the magnetic normal component B_x for mode ($m = 2, n = 1$) at $T = 1000$.

Corresponding to the decrease of the electromagnetic energy, the particles gain energy significantly, as shown in Figure 6*b*. This increase of the particle energy is much stronger than that in the cases of single tearing mode (note that the amplitudes of the high-frequency ripples with period around 50 is of the same order for both the single- and multiple-mode cases). By the end of the run at $T = 1500$, the particles have gained energy 60% more than the initial energy and the heating is still continuous. This phenomenon is completely different from the case in which only one single tearing mode exists in previous cases. The increase of the particle energy should be the same as the decrease of the electromagnetic energy for the reason of conservation of total system energy. This can be verified by Figure 6*c*, in which the increase of particle energy $(E_p(t) - E_p(0))$ and decrease of electromagnetic energy $(-(E_em(t) - E_em(0))$ coincide. In fact, the total energy is conserved within 10^{-4}. Figure 6*d* shows the time variation of the energy of the electrostatic (or longitudinal) component. The electrostatic energy (Figure 6*d*) increases as the particle energy increases, but then becomes saturated at $T \approx 1250$. Figure 6*e* shows the time variation of the magnetic helicity $H = \int_\tau \mathbf{A} \bullet \mathbf{B} dx^3$. The integration is carried out over the whole simulation box. The panel shows that as the energy increases the magnetic helicity decreases till $T = 1250$ and then increases slightly. The maximum change of the magentic helicity is about 20%.

In the following five panels (Figures 6*f*-6*j*), we show the time history of the magnetic energy for each individual mode.

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Figure 6*f* shows the time variation of $E_B(m = 0, n = 0)$, i.e., the background magnetic field energy. It decreases with time, and after $T = 750$, it is reduced significantly. This background magnetic field is the free energy source for the particle heating and the tearing mode growth. Figure 6*g* shows the magnetic energy for mode $(m = 0, n = 1)$. It shows that this mode does not grow until $T = 750$, and then grows very rapidly. The growth rate calculated for the interval from $T = 750$ to $T = 1250$ is $\gamma \approx 0.008\omega_{pe}$, which is more than 3 times bigger than that in the case of the isolated condition shown in Figure 4.

Figure 6*h* shows the magnetic energy for mode $(m = 1, n = 1)$, which develops on the lower side of the current layer. This mode grows slowly at the beginning until $T = 750$, and then grows rapidly. During the interval from $T = 750$ to $T = 1000$, it has a growth rate of $\gamma = 0.006\omega_{pe}$. The mode $(m = -1, n = 1)$ which develops on the opposite side of the current layer has similar properties to mode $(m = 1, n = 1)$. Its magnetic energy is shown in Figure 6*i*, in which the slight difference from mode $(m = 1, n = 1)$ is due to a slight asymmetry in the initialization. Figure 6*j* shows the energy plot for mode $(m = 2, n = 1)$. It has much smaller growth than those with small mode numbers during the run. The modes $(m = -2, n = 1)$, $(m = -3, n = 1)$ and $(m = 3, n = 1)$ do not grow significantly either (not shown).

In the following four panels (Figures 6*k*-6*n*), we show the snapshots of eigenmode profile for each modes, or $B_x(x, m, n)$ versus x, at $T = 1000$. In these panels, the solid and dashed traces correspond to the imaginary and real parts, respectively.

In Figure 6*k*, $B_x(x, 0, 1)$, the normal magnetic field component for mode $(m = 0, n = 1)$, is maximized at $x = 64\lambda_e$, the center of the current layer. This is slightly different from the case of the isolated tearing mode; i.e., the dip at $x = 64\lambda_e$ in Figure 4*j* disappears. In Figures 6*l* and 6*m*, B_x for modes $(m = 1, n = 1)$ and $(m = -1, n = 1)$ are peaked at each side of the current layer and evanescently decaying away from the maxima. It is worth noting that the maximum is not at the resonance surface location where $\mathbf{k} \bullet \mathbf{B} = 0$. This property is similar to the isolated tearing mode case in section 3.2.

To illustrate the particle diffusion, we show the phase space of the ions. Figure 7 shows the particle position x_i, y_i, z_i projected onto the xz plane for times $T = 250, 500, 750, 1000$, and 1250. Initially, a Harris equilibrium is loaded, and therefore there are more particles near $x = 64\lambda_e$ at $T = 250$. At time $T = 500$, no significant change on the particle distribution can be seen. The particle diffusion becomes more obvious at $T = 750$, when particles are stretched out at the left side of the current layer as a result of $(m = 1, n = 1)$ mode developing. At time $T = 1000$, particles diffuse across the current layer more quickly on both sides, though still stronger on the left side than on the right side. At time $T = 1250$, particles have nearly completely diffused across the current layer. The density becomes quite uniform inside the simulation box. Following the diffusion of particles, thecurrent layer becomes broader in width, too, although it does not diffuse as strongly as the particles.

The strong change of energies are caused by strong changes in current and particle distributions. In the isolated tearing mode case, we do not see any significant change of energies, and we do not see any significant change in particle and current distributions, either. This is because the current is flattened as soon as the tearing mode island grows to the size of the current layer width. Therefore free energy is exhausted for the tearing mode. However, this is not case for the multiple tearing modes. As one can see in Figure 8, the flattening of current on one mode will be compensated by the development of the other modes. When one mode tends to flatten the current, the development of the other mode creates a slope on it to provide more free energy.

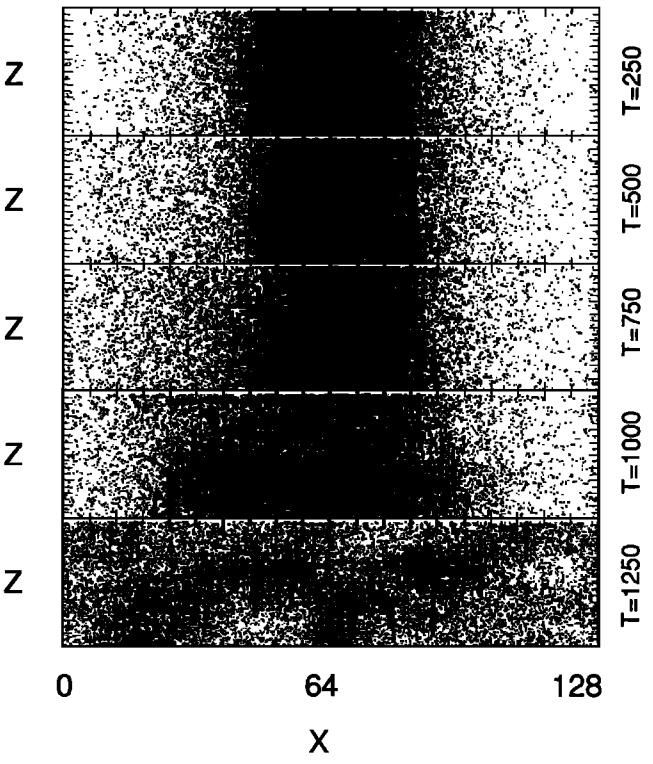


Fig. 7. Simulation results for the case with multiple modes showing snapshots of the particle phase space (ions) at different times.

In Figure 8, we show the snapshots of density, current, and magnetic field B_z profile at $T = 250, 500, 750, 1000$ and 1250. As time goes on, we can see the broadening of the current layer, the diffusion of the particles, and the flattening of B_z. The current is first flattened at the location $x = 48\lambda_e$ (Figure 8*c*) due to the development of mode $(m = 1, n = 1)$. We then see a decrease of the current value at the center of the current layer (Figure 8*d*). This is due to the development of the centered tearing mode, which causes the current diffusion at that location. At time $T = 1250$, the density is really flattened in most of the simulation box, where the current still has a peak at the center of the current layer. This tells us that the diffusion of particles is stronger than that of the current. In the next section, we will discuss the diffusion coefficient and resistivity when they are applied to the magnetopause condition.

Next we illustrate the magnetic field line topological properties through the surface of section plots (See Appendix B for detailed description). The surface of section plot for the field lines consists of intersection points of a bundle of field lines on the plane which cuts the field lines. The pattern of these intersection points tells us the topology of the field lines. For instance, if the pattern is very random, it indicates that no magnetic surface exists and the field lines are stochastic.

We examine the evolution of field line stochasticity using the surface of section plot at a series of times. In addition to

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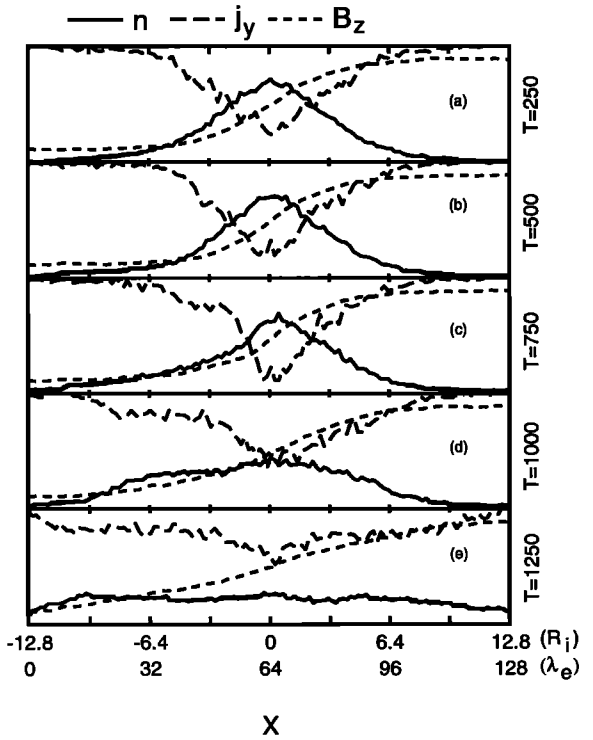


Fig. 8. Simulation results for the case with multiple modes showing snapshots of density n, magnetic field B_z component, and current j_y at different times.

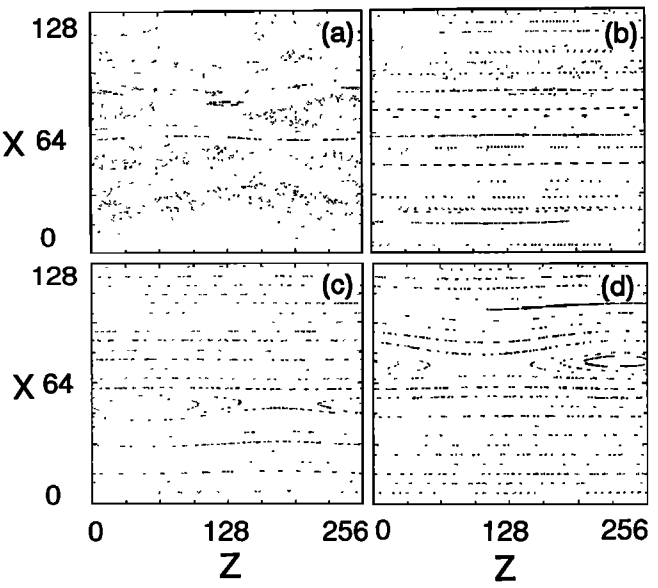


Fig. 9. Simulation results for the case with multiple modes showing the surface of section plots at $T = 250$. (*a*) The overall surface of section plot. (*b*) The surface of section plot corresponding to mode $(m = 0, n = 1)$. (*c*) The surface of section plot corresponding to mode $(m = 1, n = 1)$. (*d*) The surface of section plot corresponding to mode $(m = 1, n = -1)$.

the surface of section plot for the overall field, to determine which modes are responsible for the field line stochasticity, we also need the surface of section plot for each mode; i.e., the magnetic field is constituted by the sum of the background field and the field of a specified mode, $\mathbf{B} = \mathbf{B}(x, m, n) + \mathbf{B}(x, 0, 0)$.

Figure 9 shows the surface of section plots at $T = 250$. Figure 9*a* is the overall surface of section plot which includes contribution from all the modes. Even at this early stage, some localized field line stochasticity has developed in certain areas. However, regular field lines exist at the center of the current layer.

We now choose only a specified mode with the background field to draw the surface of section plot. Figure 9*b* shows the surface of section plot corresponding to mode $(m = 0, n = 1)$. It does not show any island structure, which means the tearing mode has not been excited yet at this early stage. This can also be seen from the mode energy in Figure 6*g*.

Figures 9*c* and 9*d* show the surface of section plots corresponding to modes $(m = 1, n = 1)$ and $(m = 1, n = -1)$, respectively. We can see that the tearing modes have developed at this time. and they are the cause of stochasticity shown in Figure 9*a*. If one carefully examines Figure 9*a*, one can still see traces of these islands. The localized stochastic regions around the current layer in Figure 10*a* are caused by the interaction of these two tearing modes, although the tearing islands in these two modes have not completely overlapped.

Figure 10 is in the same format as in Figure 9 and shows the surface of section plots at $T = 500$. In Figure 10*a* (the overall surface of section plot), field lines become stochastic where regular magnetic surfaces still exist outside the current layer. The tearing island in mode $(m = 0, n = 1)$ appears in the center of the current layer at this time (Figure 10*b*), which overlaps the islands in modes $(m = 1, n = 1)$ and $(m = 1, n = -1)$. Furthermore, the tearing islands in modes $(m = 1, n = 1)$ and $(m = 1, n = -1)$ are also enlarged (Figures 10*c* and 10*d*). The strong stochasticity around the current is due to the overlap of these islands.

The surface of section plots for $T = 1000$ are shown in Figure 11. At this time, the field line stochasticity has become

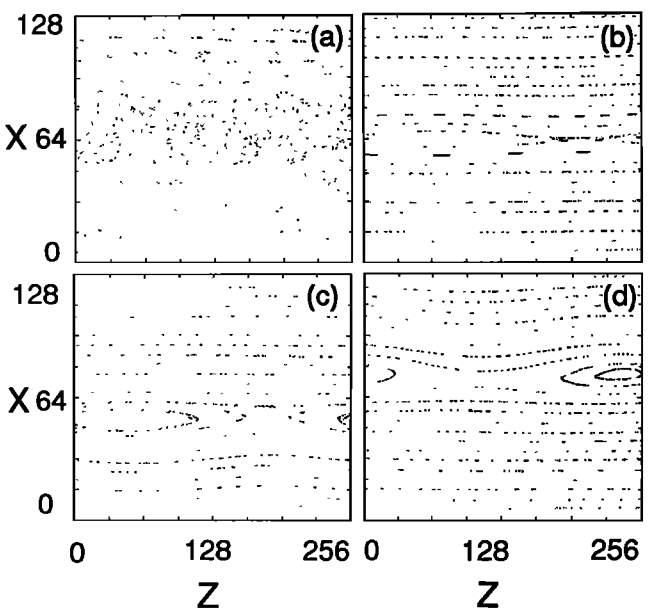


Fig. 10. Simulation results for the case with multiple modes showing the surface of section plots at $T = 500$. (*a*) The overall surface of section plot. (*b*) The surface of section plot corresponding to mode $(m = 0, n = 1)$. (*c*) The surface of section plot corresponding to mode $(m = 1, n = 1)$. (*d*) The surface of section plot corresponding to mode $(m = 1, n = -1)$.

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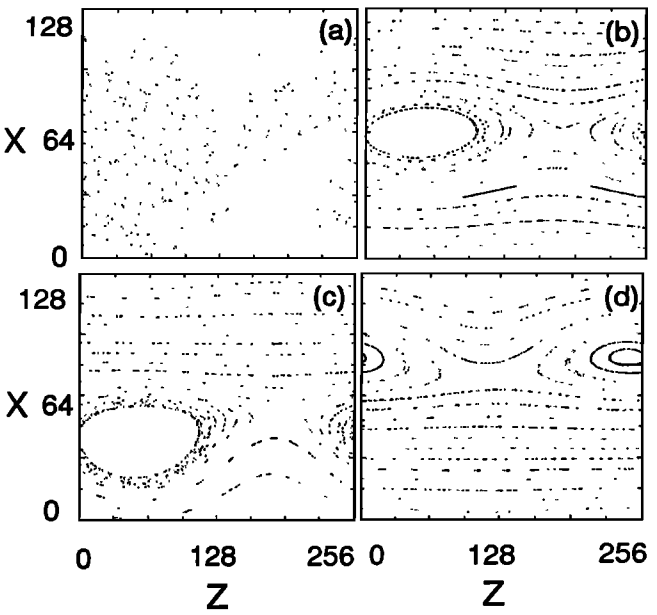


Fig. 11. Simulation results for the case with multiple modes showing the surface of section plots at $T = 1000$. (*a*) The overall surface of section plot. (*b*) The surface of section plot corresponding to mode $(m = 0, n = 1)$. (*c*) The surface of section plot corresponding to mode $(m = 1, n = 1)$. (*d*) The surface of section plot corresponding to mode $(m = 1, n = -1)$.

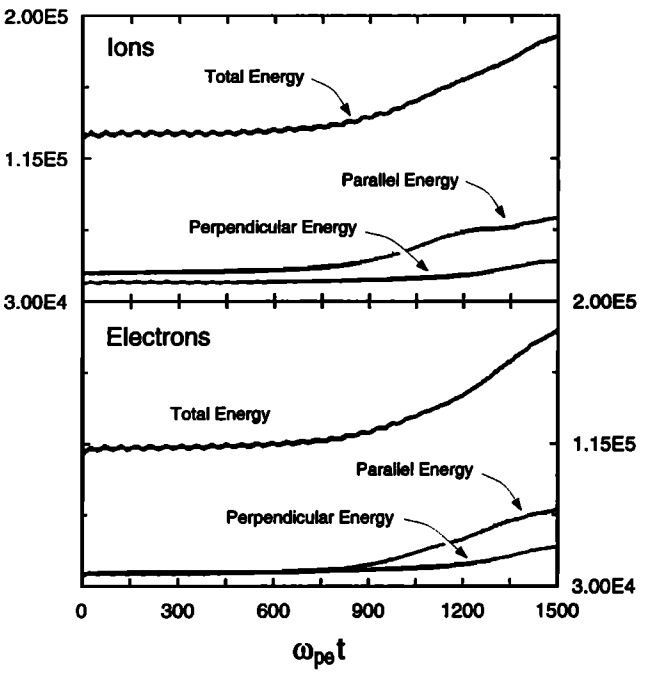


Fig. 12. Simulation results for the case with multiple modes showing the time variation of total kinetic energy, parallel energy, and perpendicular energy of (top) ions and (bottom) electrons.

global throughout the simulation system. The tearing islands in each mode (Figures 11*b*, 11*c*, and 11*d*) become so large that they strongly overlap one another. If one compares them with the islands formed by isolated single tearing mode (Figures 4*k* and 5*k*), one sees that these islands are much bigger.

As we mentioned in Figure 6*b*, the particle energy has gained 60% at the end of the simulation ($T = 1500$). The heating of the particles is in fact mainly in the direction of the local magnetic field. To show this, we have separated the particle energy into two parts, parallel and perpendicular with respect to the local magnetic field. In Figure 12, we show the time variation of particle energy, including the total kinetic energy, parallel energy, and half of the perpendicular energy. (Note there are 2 degrees of freedom in the perpendicular direction.) The upper panel is for ions, and the lower panel is for electrons. It shows that for both electrons and ions, the parallel energy increases more than the perpendicular energy. As the total particle energy begins to increase rapidly at $T = 750$, the parallel energy increases at the same rate (the slopes of these two curves are about the same). However, the perpendicular energy starts to increase at a later time $T = 1200$ and becomes increasing as fast as the parallel energy at $T = 1500$. At $T = 1500$, the parallel energy is still 50% greater than the half of the perpendicular energy. In Figure 12, we can also notice a fast oscillation due to the slight imbalance in the initial loading. This oscillation appears in the perpendicular energy but not in the parallel one. Thus the oscillation is perpendicular to the magnetic field.

4. Discussion and Conclusions

The onset of global stochasticity of magnetic field lines requires the overlap of magnetic islands generated at different magnetic surfaces inside the current layer. Such a criterion can be rephrased by a renormalized fluctuation amplitude $\epsilon > 4$ (see Appendix A), where

$$\epsilon = \frac{k_z B_{zo} \delta B_x}{k_{yo}^2 B_{yo}^2 L} 4\pi^2$$

If we use our simulation parameters in Table 1, we get $\epsilon \approx 321 \delta B_x / B_{yo}$. This means a fluctuation slightly over 1% will cause the field lines to be stochastic. Such a tearing mode fluctuation level is accomplished very quickly in the early stage of the simulation.

The field line stochasticity will produce a very strong particle diffusion across the mean magnetic field. The particle diffusion coefficient has been obtained theoretically by calculating the magnetic field line diffusion coefficient D_m, first using the quasi-linear theory [*Rosenbluth et al.*, 1966], and then multiplying by the ion thermal speed, i.e., $D_p = V_T \cdot D_m$ [e.g., *White*, 1983]. In our numerical simulation, we only have a discrete spectrum of modes with limited mode number. However, if the islands of these modes are strongly overlapped, the diffusion is in the quasi-linear limit [*Rechester and White*, 1980]. In this case, we have

$$D_m = \frac{\pi}{2} \frac{1}{k_{yo}} \frac{\delta B_x^2}{B_{yo}^2}.$$

The particle diffusion coefficient will then be

$$D_p = D_{renorm} \frac{V_T^2}{\Omega_c},$$

where

$$D_{renorm} = \frac{L_y / R_i}{4} \frac{\delta B_x^2}{B_{yo}^2},$$

L_y is the fundamental mode wavelength in the y direction, and

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R_i is the ion Larmor radius. In our simulation, the magnetic field fluctuation level is changing with time. If we take a reasonable amplitude from our simulation $\delta B_x/B_{yo} \sim 0.1$, the renormalized diffusion coefficient from the quasi-linear theory is $D_{renorm} \sim 0.25$. We can also calculate the diffusion coefficient directly from the density profile in Figures 8*a* to 8*e*. Let us assume the density has a Gaussian distribution, i.e.,

$$n(x,t) = \frac{N}{\sqrt{\pi} l(t)} \exp(-\frac{x^2}{l^2(t)})$$

$$l^2(t) = D(t + t_o)$$

where D is the diffusion coefficient and t_o is a free parameter to give the initial scale length L. Although our initial density profile is not a Gaussian, we still treat it as a Gaussian since $\mathrm{sech}^2(x/L)$ is similar to $\exp(-x^2/L^2)$. We therefore can get the diffusion coefficient from $D = [l^2(t) - L^2]/t$. The renormalized diffusion coefficient is

$$D = \frac{(L/R_i)^2 (n^2(x=0,t=0)/n^2(x=0,t) - 1)}{\Omega_c t}$$

After taking the results from Figures 8*a* to 8*e*, we get $D \sim 0.24$, which is of the same order as the quasi-linear theory result. The value of V_T^2/Ω_c is about 10^{10} m^2/s for typical magnetopause parameters$R_i = 40$ km and $\Omega_c = 2\pi$ s^{-1}. We therefore have a diffusion coefficient of the order of 10^9 m^2/s due to the stochastic field line onset at the magnetopause. This diffusion coefficient is bigger than the ones obtained by microinstabilities [*LaBelle and Treumann*, 1988].

Let us now look at the resistivity by examining the diffusion of the current. The particle diffusion across the mean magnetic field should in principle produce a current diffusion, since the current is carried by particles. However, the diffusion in current will reduce the magnetic energy, which in turn heats the particles. This means the diffusion in current has to be accompanied by a velocity space diffusion in particles. Such a diffusion process in velocity has to be coupled with the diffusion in real configuration space. We have seen in our simulation that as the particles diffuse across the current layer, they are heated due to the dissipation of magnetic field energy. The current diffusion coefficient (or resistivity) estimated from the change of the scale length of the current profile is

$$\eta = \frac{(l_c^2(t) - L^2)/R_c^2}{\Omega_c t} (\frac{V_T^2}{\Omega_c}) \approx 0.1 (\frac{V_T^2}{\Omega_c})$$

which is about 2-3 times smaller than the particle diffusion coefficient.

The relaxation of the system after field line stochasticity is certainly toward minimizing the magnetic energy. Such minimizing of magnetic energy has been considered to be under a constraint; i.e., the total magnetic helicity $H = \int_\tau \mathbf{A}\cdot\mathbf{B} dx^3$ is conserved [*Taylor*, 1974]. With such a constraint, the system is evolving to a force-free state when the magnetic energy of the systems reduces to its minimum, the so-called Taylor state. In fact, the magnetic helicity is a rigorous invariant only in ideal MHD and under the condition that the integral is taken in a volume τ confined by magnetic surfaces. When magnetic surfaces do not exist any more as the magnetic field lines become stochastic, *Taylor* [1974] argued that the magnetic helicity, which is integrated over the whole space, should still be a good invariant because it summed over all of the flux tubes. We have shown in Figure 6*e* that the magnetic helicity $H = \int_\tau \mathbf{A}\mathbf{dot}\mathbf{B} dx^3$, where the integral over the simulation box is changing with time. When comparing with the energy change in particles, we find that when the particle energy increases 60%, the magnetic helicity changes by about 20%. Thus the helicity is not a very strict invariant in our stochastic field line system.

From our numerical simulation, we draw the following conclusions: The magnetic field lines can become stochastic inside the magnetic field current layer due to the growth and the overlap of the multiple tearing mode islands, which agrees with the percolation model suggested by *Galeev et al.* [1986]. After the field lines become stochastic, the tearing modes which cause the stochasticity grow 2-3 times faster and the saturation amplitude becomes 3-4 times greater than that in the linear stage. The particle diffusion coefficient reaches to 10^9 m^2/s when typical magnetopause parameters are used. Particles are strongly heated as the current layer broadens due to the field line stochasticity. The relaxation of the system is toward minimizing the magnetic energy, but the conservation of magnetic helicity is not strict.

Appendix A: Kinematic Study of the Field Line Stochasticity

In this section, we present a brief discussion of field line stochasticity using kinematic arguments for a Harris current sheet.

For a modified Harris equilibrium, the background magnetic field is $B_z = B_{zo}\tanh(x/L)$ and $B_y = B_{yo}$. The fluctuating normal component B_x from the tearing modes has the form

$$B_x = \sum_{m,n} B_x(x,m,n) e^{\imath m k_{zo} z + \imath n k_{yo} y} \tag{17}$$

If we trace field lines using this magnetic field, we will know some of its topological properties. We now simplify the magnetic field slightly and make an analytical analysis on this set of magnetic field. First we assume that the background magnetic field is not strongly changed at the stochasticity onset. Second, we assume that the $n = 0$ mode does not grow, because it is not a tearing mode. Third, we assume that the mode amplitude does not change in the x direction and independent of the mode number, i.e., $B_x(x,m,n) \equiv \delta B_x \equiv$ const. The third assumption is purely for the sake of simplification. We then have the following equations for the field lines:

$$\frac{dx}{dy} = \frac{B_x}{B_y} = \frac{\delta B_x}{B_{yo}} \sum_{n=-N}^{N} \sin(k_{zo} z + n k_{yo} y) \tag{18}$$

$$\frac{dz}{dy} = \frac{B_z}{B_y} = \frac{B_{zo}}{B_{yo}} tanh(\frac{x}{L}) \tag{19}$$

The stochastic property should in principle exist in this set of equations. To make these equations more tractable for analytical analysis, we renormalize the equations by using

$$\xi = \frac{x}{L}\beta, \quad \zeta = k_{zo} z, \quad s = \frac{k_{yo}}{2\pi} y$$

$$\beta = \frac{k_z B_{zo}}{k_{yo} B_{yo}} 2\pi, \quad \epsilon = \frac{k_z B_{zo} \delta B_x}{k_{yo}^2 B_{yo}^2 L} 4\pi^2$$

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and the renormalized positions x, z, and y, i.e., ξ, ζ, and s satisfy the following equations:

$$\frac{d\xi}{ds} = \epsilon \sum_{n=-N}^{N} sin(\zeta - 2\pi n s) \tag{20}$$

$$\frac{d\zeta}{ds} = \beta tanh(\frac{\xi}{\beta}). \tag{21}$$

As in most stochastic system analyses [e.g., *Lichtenberg and Liberman*, 1983], we will obtain a mapping equation for (20) and (21). The mapping equation describes the relation between one point on the surface of section plot (shown in the last section) and the next point. That is, we have integrated equations (20) and (21) in s (y before renormalization). To perform an analytic integral on (20) and (21) is not easy. However, if we set maximum mode number $N \to \infty$, equations (20) and (21) become

$$\frac{d\xi}{ds} = \epsilon \sin(\zeta) \sum_{n=-\infty}^{\infty} \delta(s - n) \tag{22}$$

$$\frac{d\zeta}{ds} = \beta \tanh(\frac{\xi}{\beta}) \tag{23}$$

If we integrate the last two equations from n to $n+1$, we get the mapping equations

$$\xi^{n+1} = \xi^n + \epsilon \sin(\zeta^n) \tag{24}$$

$$\zeta^{n+1} = \zeta^n + \beta \tanh(\frac{1}{\beta}\xi^{n+1}) \tag{25}$$

The mapping equations are different from the standard mapping equations [*Chirikov*, 1979] in the second term of the RHS of (25). But when $\xi \ll \beta$, they reduce to the standard mapping equations. We now apply linear stability analysis to the mapping equations. Suppose that at step n, ξ^n and ζ^n deviate by ϵ_ξ^n and ϵ_ζ^n, respectively. This deviation will propagate to step $n+1$ by (24) and (25), and the deviation at $n+1$ will satisfy the following equation:

$$\begin{pmatrix} \epsilon_\xi^{n+1} \\ \epsilon_\zeta^{n+1} \end{pmatrix} = \begin{pmatrix} 1 & \epsilon cos(\zeta^n) \\ \text{sech}^2(\frac{\xi^{n+1}}{\beta}) & 1 + \text{sech}^2(\frac{\xi^{n+1}}{\beta})\epsilon\cos(\xi^n) \end{pmatrix} \begin{pmatrix} \epsilon_\xi^n \\ \epsilon_\zeta^n \end{pmatrix} \tag{26}$$

If the deviation at $n+1$ is larger than that at n, the equations are unstable. Therefore we need to determine the eigenvalue of the following equations:

$$\begin{pmatrix} \epsilon_\xi^{n+1} \\ \epsilon_\zeta^{n+1} \end{pmatrix} = \lambda \begin{pmatrix} \epsilon_\xi^n \\ \epsilon_\zeta^n \end{pmatrix} \tag{27}$$

Using (26) and (27), we obtain

$$\lambda^2 - (2 + \text{sech}^2(\frac{1}{\beta}\xi^{n+1})\epsilon\cos(\zeta^n))\lambda + 1 = 0 \tag{28}$$

which gives the solution

$$\lambda = \frac{b \pm \sqrt{b^2 - 4}}{2}$$

where

$$b = 2 + \text{sech}^2(\frac{1}{\beta}\xi^{n+1})\epsilon\cos(\zeta^n)$$

If $|b| > 2$, $|\lambda|$ has a solution larger than unity, and equations (24) and (25) are unstable. If $|b| \leq 2$, then the equations are stable. For $\zeta \sim \pi$, if $\epsilon < 4$, we have $|b| < 2$; the field lines around that point are regular. If $\epsilon > 4$, however, stability will depend on the value of x^{n+1}. If $x^{n+1} \sim 0$, it is unstable; if $x^{n+1} \gg 1$, it is stable. If $x^{n+1} \to \infty$, then $\lambda \to 1$, and therefore (24) and (25) will be marginally stable. The field lines inside the current layer are more easily made stochastic than those outside the current layer.

Since at $\xi \sim 0$ the mapping equation reduces to the standard mapping equation, we know that the diffusion coefficient at $\xi \sim 0$ is approximately the quasi-linear diffusion value [*Rechester and White*, 1980], i.e.,

$$D_{norm} \equiv \frac{< (\xi - \xi_o)^2 >}{s} \approx \frac{\epsilon^2}{4} \tag{29}$$

where

$$D_{norm} = \{\frac{1}{L}\frac{k_z B_{zo}}{k_{yo} B_{yo}} 2\pi\}^2 \frac{2\pi}{k_{yo}} \frac{< (x - xo)^2 >}{y}$$

Hence

$$D_m \equiv \frac{< (x - xo)^2 >}{y} = \frac{\pi}{2}\frac{1}{k_{yo}}\frac{\delta B_x^2}{B_{yo}^2}$$

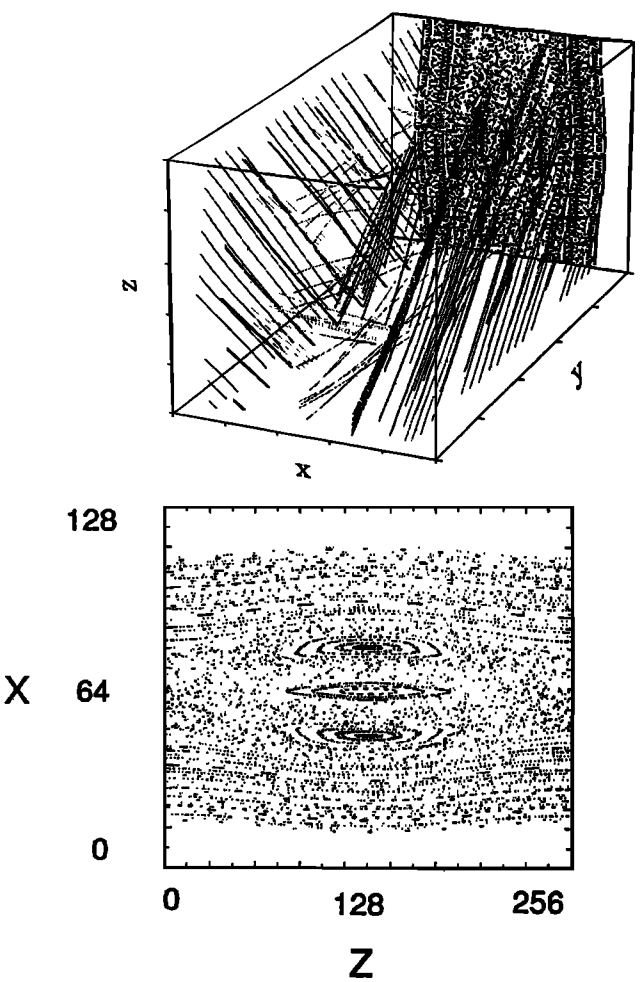


Fig. B1. Schematic for the construction of the surface of section plot for the magnetic field lines at one time. (Top) Magnetic field lines in the simulation box. (Bottom) Corresponding surface of section plot in the xz plane at $y = L_y$.

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Because the particles are moving along the stochastic field lines, the particle diffusion coefficient is then

$$D_p = V_T D_m \tag{30}$$

Appendix B: Surface of Section Plot

The surface of section plot in Figure B1 is generated in the following manner. For a given time, we start tracing a field line using the magnetic field value given at that time. As the field line intersects with the xz plane at $y = L_y$, we draw a dot at the intersection point. We then continue tracing this field line but from the xz plane at $y = 0$, since the system is periodic in the y direction. While tracing this field line, it may reach $z = L_z$ too. Since the system is periodic in the z direction, we will continue tracing it from $z = 0$ with the same x and y values. Our system is also limited in the x direction by L_x. If the field line reaches the $x = L_x$ plane, we stop tracing this field line. We also limit our tracing by certain steps. After we finish tracing this field line, we will have several dots on the xz plane at $y = L_y$. We then repeat this procedure for another field line with slightly different initial location, and so on. We choose to trace 100 field lines. The final surface of section plot is shown in the bottom part Of Figure B1. . From this figure, it is easy to see that in some areas field lines are regular, e.g., the circular pattern around $x = 64$ and $z = 128$ and the linear pattern at large x. We can also see that in some areas field line are stochastic where the dots are randomly distributed, e.g., around $x = 64$ and $z = 0$.

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